100 SQL Lab Exercises

DML Commands, Queries, Aggregate Functions and Table Structure

Database: PostgreSQL **Level:** Beginner to Intermediate

Instructions

1. Complete the exercises in order.
2. Use the **contacts** and **products** tables from your classroom notes.
3. For UPDATE and DELETE exercises, verify the target rows with SELECT before changing data.
4. PostgreSQL uses single quotes for text values, for example `WHERE category = 'Electronics'`.
5. Some exercises change table structure or data. Use a fresh practice database or recreate the tables when needed.

Section 1: INSERT Operations (1–10)

1. Insert a new contact with slno 21, name 'Meera', email 'meera@gmail.com', and mobile 9876501234.

2. Insert five new contacts in a single INSERT statement. Use slno values 22 to 26.

3. Insert a contact using only slno and name. Observe what happens to the remaining columns.

4. Insert a contact with a NULL email address.

5. Insert a contact whose name is 'Mohan Kumar' and email is 'mohan.kumar@gmail.com'.

6. Insert three products in one INSERT statement. Include product name, category, brand, price, and quantity.

7. Insert a new product in the 'Electronics' category with a price greater than 30000.

8. Insert a product with quantity 0.

9. Insert a product whose brand contains two words, such as 'Dell Technologies'.

10. Insert two contacts with different names but the same email domain.

Section 2: SELECT, Aliases, LIMIT and ORDER BY (11–20)

11. Display all columns and all rows from the contacts table.

12. Display only the name and email columns from contacts.

13. Display product_name, category, and price from products.

14. Display each contact's name as customer_name and mobile as contact_number.

15. Display product_name as item and price as selling_price.

16. Display the first five records from contacts.

17. Display the first seven products from products.

18. Display all products ordered by price in ascending order.

19. Display all products ordered by price in descending order.

20. Display products ordered first by category in ascending order and then by price in descending order.

Section 3: WHERE and Relational Operators (21–32)

21. Display contacts whose slno is greater than 10.

22. Display contacts whose slno is less than 8.

23. Display the contact whose slno is exactly 15.

24. Display contacts whose slno is not equal to 10.

25. Display products with price greater than 5000.

26. Display products with price less than or equal to 1000.

27. Display products whose quantity is greater than 30.

28. Display products whose price is exactly 180.

29. Display products whose quantity is not equal to 15.

30. Display products with price greater than 1000 and quantity greater than 20.

31. Display products with price less than 1000 or quantity greater than 70.

32. Display products whose category is not 'Clothing'.

Section 4: Logical Operators and BETWEEN (33–42)

33. Display contacts whose slno is between 5 and 12 using AND.

34. Display contacts whose slno is between 5 and 12 using BETWEEN.

35. Display products priced between 1000 and 5000.

36. Display products with quantity between 10 and 30.

37. Display products in the Electronics category with price greater than 10000.

38. Display products in the Clothing or Footwear categories.

39. Display products that are not in the Groceries category.

40. Display products where price is greater than 1000 and less than 10000.

41. Display products where quantity is below 20 or price is above 20000.

42. Display products that are in Electronics and have quantity greater than 10, or are in Footwear.

Section 5: LIKE Pattern Matching (43–52)

43. Display contacts whose names start with the letter 'A'.

44. Display contacts whose names end with the letter 'n'.

45. Display contacts whose names contain 'an'.

46. Display contacts whose names start with 'M' and have any remaining characters.

47. Display products whose product_name starts with 'S'.

48. Display products whose product_name contains the word 'Shoes'.

49. Display products whose brand contains 'Samsung'.

50. Display products whose category ends with 'wear'.

51. Display contacts whose email addresses end with 'gmail.com'.

52. Display contacts whose names have exactly four characters using the appropriate LIKE wildcard.

Section 6: IN and Subqueries (53–62)

53. Display contacts whose slno is in (1, 2, 3, 4, 5).

54. Display contacts whose slno is in (3, 7, 11, 15, 19).

55. Display products belonging to Electronics, Clothing, or Footwear using IN.

56. Display products whose brand is Samsung, Puma, or Nike.

57. Display products whose price is in the set (50, 100, 180, 800).

58. Use a subquery to display contacts whose slno is less than 10.

59. Use a subquery to display products whose price equals the maximum product price.

60. Use a subquery to display products whose price equals the minimum product price.

61. Use a subquery to display products whose quantity is greater than the average quantity.

62. Use a subquery to display products belonging to the same category as the product named 'Laptop'.

Section 7: GROUP BY, HAVING and Aggregate Functions (63–76)

63. Count the total number of contacts.

64. Count the total number of products.

65. Find the total quantity of all products using SUM.

66. Find the average product price using AVG.

67. Find the highest product price using MAX.

68. Find the lowest product price using MIN.

69. Display the number of products in each category.

70. Display the total quantity available in each category.

71. Display the average price for each category.

72. Display the maximum price in each category.

73. Display the minimum price in each category.

74. Display categories having more than three products using HAVING.

75. Display categories whose total quantity is greater than 100 using HAVING.

76. Display category-wise product counts in descending order of the count.

Section 8: UPDATE and DELETE (77–86)

77. Change the price of the product whose product_id is 13 to 1400.

78. Increase the price of every Electronics product by 5 percent.

79. Set the quantity of the product named 'Mouse' to 60.

80. Change the brand of the product named 'T-Shirt' to 'Puma Sports'.

81. Update the category of the product named 'Sandals' to 'Footwear'.

82. Reduce the quantity of all products in the Groceries category by 5.

83. Update a contact's email address using its slno.

84. Set the price to 0 for a product identified by its product_id (use a suitable test record).

85. Delete one contact using its slno.

86. Delete products whose quantity is 0. Check the affected rows before and after deletion.

Section 9: ALTER TABLE Operations (87–94)

87. Add a column named stock of type INTEGER to products.

88. Add a column named description of type VARCHAR(250) to products.

89. Rename the quantity column to qty.

90. Change the datatype of qty to DECIMAL(10,2) in PostgreSQL.

91. Change the datatype of brand to VARCHAR(100) in PostgreSQL.

92. Drop the description column from products.

93. Rename the contacts table to customers.

94. Add a column named created_on of type DATE to the customers table.

Section 10: Integrated Practice (95–100)

95. Create a report showing each category, number of products, total quantity, and average price. Sort by average price from highest to lowest.

96. Display categories whose average product price is greater than 5000.

97. Find the most expensive product in each category. Use GROUP BY and a subquery or another suitable query.

98. Display products that cost more than the average price of all products and sort them from highest to lowest.

99. Create a low-stock report showing products with quantity below 20, displaying product name, category, quantity, and price.

100. Perform a complete practice sequence: insert a product, select it, update its price and quantity, verify the changes, and finally delete that product.